

Algebraic Structures - Coursework 1

- (i) The questions **are not organised by difficulty**.
- (ii) Submit your solutions on Blackboard. The deadline for submission is **Monday, 28 October at 4pm**.
- (iii) Although you are allowed to submit your solutions after the deadline, if you do, you will be penalised with a mark reduction according to the University's rules.
- (iv) Write your solutions on paper sheets, then scan them and submit through Turnitin. If you have no scanner available, you may also take photos on your phone, but note that you may have to bundle those together in a **single file for uploading**. Writing your solutions on a tablet is also allowed.
- (v) **Justify all your answers and write down all the steps of your calculations.**
- (vi) Submit only one version of each solution (cross out wrong attempts).

Question C1.1

In this question, we consider the multiplicative group

$$\mathrm{GL}(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R}, \text{ and } ad - bc \neq 0 \right\}$$

and its subset

$$H = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in \mathbb{R}, \text{ and } ad = 1 \right\}.$$

- (a) Use one of the criteria we learnt in class to show that the set H defined above is a subgroup of $\mathrm{GL}(2, \mathbb{R})$ with matrix multiplication. [15 marks]

- (b) Let $P = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \in H$. Show that the following map is a group homomorphism. Justify your answer. [10 marks]

$$f : H \rightarrow H \quad \text{given by } f(A) = P^{-1}AP, \quad \text{for each } A \in H.$$

- (c) Is the map $f : H \rightarrow H$ from item (b) also a group isomorphism? Justify your answer. [10 marks]

- (d) Is H an abelian subgroup? Justify your answer. [10 marks]

- (e) Check whether the subgroup [10 marks]

$$K = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in \mathbb{R} \right\}$$

is a normal subgroup of H . Justify your answer.

[You do not need to show that this is a subgroup, only the condition for being normal.]

Question C1.2

In this question, we consider the group $\mathbb{Z}_{12}$ of integers modulo 12.

(a) Find the order of the element $\overline{4}$. Justify your answer. [10 marks]

(b) Write down all different elements from $\{\overline{0}, \overline{1}, \dots, \overline{11}\}$ contained in the following subgroups of $\mathbb{Z}_{12}$: [20 marks]

(i) $\langle \overline{2} \rangle$,

(ii) $\langle \overline{6} \rangle$.

Show all steps of your computations.

(c) Show that the map $h : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{12}$ given by $h(x) = \overline{2}x$ is a group homomorphism and find its kernel. Justify your answers. [15 marks]